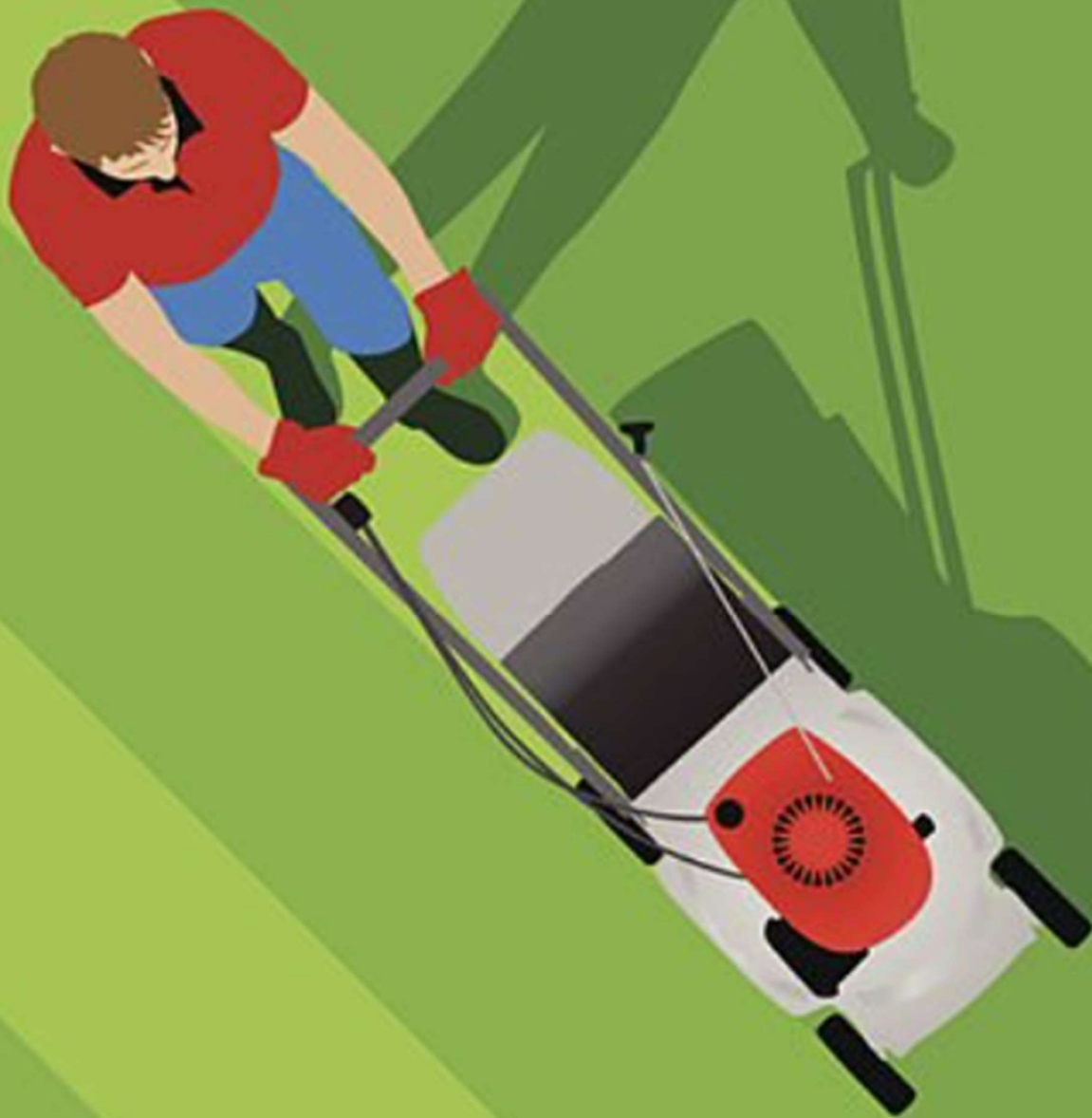


LAWN MOWER SERVICE GUIDE

ALL IMAGES ARE ROYALTY-FREE



A SERVICE MANUAL

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PRIMARY PARTS OF YOUR LAWN MOWER

DISCLAIMER

Below are the primary components of the lawn mower that will be used for this guide. Please note that this is not an exhaustive list, as different types of lawn mower (push mower, riding mower, zero-turn mower) operate on different principles and will include or exclude different parts. For example, your lawn mower specifically may be fuel injected and therefore will not have a carburetor. The purpose of this service guide is to name the key elements that make any engine run properly, and attempt to diagnose most “no-start” scenarios. When in doubt, always refer to the Owner’s manual of your lawn mower.

FUEL ASSEMBLY

FUEL CAP

A plastic cap that is found at the top of the lawn mower engine. The fuel cap prevents dirt, grass clippings, and debris from entering the fuel tank.

FUEL TANK

A plastic container found behind the lawn mower engine. The fuel tank supplies gasoline to the carburetor via a fuel hose.

FUEL HOSE

A short, rubber hose rated to supply gasoline to the carburetor.

FUEL FILTER

A fine mesh filter found inside the fuel hose, designed to prevent debris in the fuel tank from entering the carburetor. Some lawn mowers do not include a fuel filter.

FUEL SHUTOFF

A plastic valve between the fuel tank and the carburetor. One valve position will read “OFF”, obstructing the flow of gasoline.

AIR CLEANER ASSEMBLY

CARBEURATOR

A machined part that uses a combination of precisely drilled holes and a small butterfly valve to control the ratio of fuel and air going into the engine. This fuel-to-air ratio is of critical importance to engine performance and is typically one of the first components checked in troubleshooting.

CHOKE ASSEMBLY

A small lever above the air filter box. This lever operates the valve inside the carburetor, controlling the amount of air going into the engine. The lever in the “Choke” position causes the engine to run rich for a brief period. Setting the choke is useful, and often necessary when the engine is cold.

CARBEURATOR GASKET / AIR CLEANER GASKET

A series of paper-thin gaskets that seal the carburetor and prevent the intrusion of outside air into the engine.

AIR FILTER

A rectangular filter that prevents the intrusion of dirt and other debris from entering the engine. The air filter is located in the air filter box; a rectangular plastic box found on the side of the lawn mower engine.

SPARK PLUG

An electrical component made of high-grade ceramic material. It receives a large amount of voltage from the ignition coil and delivers an electric arc to the air/fuel mixture.

PULL CORD

A thin rope that is attached to a handle. It is principally used in starting the engine.

FLYWHEEL

A heavy metal wheel attached to the engine crankshaft. In a recoil-starter system, engaging the pull cord spins the flywheel and generates a magnetic force, which delivers the electrical shock to the spark plug.

BAIL BRAKE

A lever that is the same shape as the lawn mower handle. Pull this lever, holding it against the handle before engaging the pull cord.

PISTON RINGS

Small gaskets that form around each piston and create compression.

VALVES

A small, cylindrical part that acts as a door that “opens” and “closes” in time with the engine’s fuel intake, compression, combustion, and exhaust strokes.

HEAD GASKET

The gasket between the engine block and cylinder heads, which maintains compression.

SAFETY FEATURES OF YOUR LAWN MOWER

BEFORE STARTUP OR MAINTENANCE

SAFE OPERATION

KEEP HANDS AND FEET AWAY FROM MOWER DECK WHEN ENGINE IS RUNNING

KEEP MOWING AREA CLEAR OF ROCKS, WIRE, STICKS, AND TOYS THAT COULD BE THROWN BY LAWN MOWER BLADE

NEVER OPERATE LAWN MOWER WHILE BAREFOOT

DO NOT REFUEL LAWN MOWER INDOORS

STORE GASOLINE IN A COOL, DRY ENVIRONMENT AND IN APPROVED CONTAINERS.

REAR MULCH PLUG

A “plug” with a handle and 2 locking pins that is used in place of a bag. If either this plug, or lawn mower bag is missing, the lawn mower will not start as a precaution. If fuel, air, spark, and compression are all present, and it still refuses to turn? Double-check that the mulch plug is correctly installed.

BLADE ASSEMBLY

To prevent damage to the engine, objects caught inside the mower deck will cause the engine to shut off. If the pull cord feels “heavier” than normal, check that the underside of the mower deck is free of debris such as wire or dried clumps of grass.

STARTUP PROCEDURE

-
- 1)** Verify that the lawn mower is on a level surface ☐
 - 2)** Check that the rear mulch plug or bag is installed ☐
 - 3)** Check that the blade assembly is free of debris ☐
 - 4)** Check that grass discharge is free of debris ☐
 - 5)** Check fuel level ☐
-



SOME ENGINES (SUCH AS 2-STROKE) REQUIRE A FUEL/OIL MIX. SEE OWNER'S MANUAL FOR EXACT FUEL MIXTURE.

INCORRECT FUEL CAN LEAD TO ENGINE DAMAGE

-
- 6)** Check that fuel shutoff is set to “flow” ☐
 - 7)** Set choke to full, or 100% ☐
 - 8)** Set bail brake ☐
 - 9)** Engage pull cord ☐
 - 10)** Disengage choke ☐
-



SOME ENGINES MAY INCLUDE AN “AUTO-CHOKE”

TROUBLESHOOTING

ENGINE OIL CONDITION

Engine oil can degrade over time, leading to poor performance and can prevent the engine from starting at all. Change engine oil with the manufacturer's recommendation after every mowing season.



SOME ENGINES (SUCH AS 2-STROKE) REQUIRE A FUEL/OIL MIXTURE, AND DO NOT HAVE AN OIL PAN

AIR MIXTURE (TOO RICH)

An engine that is running too rich does not have enough air in the fuel mixture. Incorrect air/fuel mixture can prevent the engine from starting at all.

Symptoms of poor air mixture include

- "Popping", "Sputtering", or "Crackling"
- Excessive smoke from exhaust

AIR FILTER

Replace if the air filter is dirty or in poor condition.

CHOKE ASSEMBLY

Verify that choke assembly actuates the valve in the carburetor. The return spring may need to be reattached.

CARBURETOR

Carburetor may become clogged with dirt and debris, leading to incorrect fuel/air mixture. Remove carburetor; take care to inspect gaskets as you do so.

In most cases, thorough cleaning with carburetor cleaner will break up the debris. Once completely dry, reinstall and follow the startup procedure.

In some cases, cleaning will not be enough. The carburetor and all gaskets will need to be replaced.

FUEL MIXTURE (TOO LEAN)

An engine that is running too lean does not have enough fuel in the air mixture. Incorrect air/fuel mixture can prevent the engine from starting at all.

Symptoms of poor fuel mixture include

- “Knocking” sound like a spray can.
- It can include a high-pitched whine.

FUEL FILTER

Replace if the fuel filter is dirty or in poor condition. Some models may not include a fuel filter.

CHOKE ASSEMBLY

Verify that the choke assembly actuates the valve in the carburetor. The return spring may need to be reattached.

CARBURETOR

The carburetor may become clogged with dirt and debris, leading to incorrect fuel/air mixture. Remove carburetor; take care to inspect gaskets as you do so.

In most cases, thorough cleaning with carburetor cleaner will break up the debris. Once completely dry, reinstall and follow startup procedure.

In some cases, cleaning will not be enough. The carburetor and all gaskets will need to be replaced.

SPARK

Remove spark plug and inspect electrode. It should be dry and light tan or gray in color. If the spark plug is wet or smells of gasoline, it is “fouled” and will need to be replaced. Otherwise, it can be cleaned and reused.

COMPRESSION

If the engine has fresh oil, fresh gasoline, correct air/fuel mixture, and a working spark plug, then poor compression is likely preventing the engine from starting.

Further symptoms of poor compression include.

- Pull cord has no resistance.
- Engine previously had poor performance.
- Engine previously spun too fast.

PISTON RINGS

Small gaskets that form around each piston and create compression. If piston rings are cracked or brittle, they will need to be replaced.

HEAD GASKET

The gasket between the engine block and cylinder heads, which maintains compression. Replace if fuel seems to be leaking out of the engine block, or if the head gasket is cracked or brittle.

VALVES

Burned debris/rust can cause valves to become stuck in place. In some cases, the debris can be cleaned. In many cases, professional service may be required.

Drain all fluids and remove engine from lawn mower. Disassemble engine for access to the valves, clean debris, and reassemble engine.

IF THE LAWN MOWER STILL WILL NOT START, A FAULT INSIDE THE ENGINE BLOCK IS LIKELY PREVENTING THE ENGINE FROM GETTING COMPRESSION, AND WILL REQUIRE PROFESSIONAL SERVICE.

NOTES/MAINTENANCE LOG

[illegible]

FLOW CHART

